

1. Presentation-quality graphs — status check

Playbook required 4 graphs: UTCI timeline, nighttime UTCI vs baseline, NCTL graph, CTBI graph.

Graph	Status
UTCI timeline (6-year overview + May 2024 detail)	✅ Already existed (proof_of_work/utci_6year_overview.png, utci_may2024_timeline.png)
NCTL graph	✅ Already existed (proof_of_work/nctl_graph.png)
CTBI graph (+ sensitivity)	✅ Already existed (proof_of_work/ctbi_graph.png, ctbi_sensitivity_graph.png)
Nighttime UTCI vs baseline	NEW Was missing — built today (proof_of_work/nighttime_utci_vs_baseline.png)

The new graph shows city-averaged hourly UTCI for 18–25 May 2024 (a real peak-heat week), nighttime hours shaded, and the 90th-percentile nighttime baseline (32.11°C) drawn as a reference line — with real dates, units (°C), and a legend. Directly usable in the PPT.

2. CTBI Normalization Audit — done, real finding

Independently checked Dev's min-max normalization and the CTBI recurrence formula against the actual CTBI.csv output (not just reading the code).

Confirmed correct:

- DayStress_norm and NCTL_norm both properly bounded to [0, 1] ✅
- Both series actually span the full range (hit 0 and 1 at their real extremes) ✅
- Divide-by-zero guard present for degenerate series ✅
- CTBI itself stays within a sensible [0.094, 0.768] range for the default $\alpha=0.70$ ✅

One genuine finding for the Decision Log (not a bug, but worth documenting): The recurrence $CTBI(t) = \alpha \cdot CTBI(t-1) + \beta \cdot DayStress_norm + \gamma \cdot NCTL_norm$ uses $\beta=\gamma=0.15$. This means **$\alpha+\beta+\gamma$ only equals exactly 1.0 when $\alpha=0.70$** (the locked default). At $\alpha=0.60$ the weights sum to 0.90 (slight downward drift), at $\alpha=0.80$ they sum to 1.10 (slight upward drift). This doesn't invalidate the sensitivity test — the playbook only requires the *qualitative pattern* (rise-to-peak, decline-after) to stay stable across α , which Khushi's

independent sanity check already confirmed (NCTL $\leftrightarrow$ peak-CTBI correlation = 0.944, all 6 seasons pass). But $\alpha=0.6/0.8$ CTBI values are **not on an identical scale** to $\alpha=0.7$ — good to have one line about this ready for Q&A ("why 0.7?" / "what does the sensitivity test actually show?").